

Philips Medical Systems

www.medical.philips.com/us

PHILIPS

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Electrical Requirements
ACQSIM CT System

Supply Configuration: 3 phase, 3 wire power and ground.

Nominal Line Voltage: 480 VAC + / - 5%, 50 / 60 Hz

Branch Power Requirement: 75 KVA

Circuit Breaker: 3 pole, 100 Amps

(02.0)

HVAC Requirement for General Equipment Locations

Operating temperature range throughout the CT suite is 68 ° F. (20 ° C.) to 75 ° F. (24 °C.) at 30% to 60% relative humidity (non-condensing). Change rate 5 ° F / hr.

Power conditioning will accept ambient temperature range of 32 ° F. (0 ° C.) to 104 ° F. (40 ° C.).

Heat output in one area of CT suite must not affect temperature and humidity in other areas. It is strongly recommended that any definable areas within the suite, i.e.: equipment closets, control areas, etc. (if applicable), be individually environmentally controlled as required to meet the ambient ranges specified.

See equipment planning code for individual equipment ratings which may or may not operate simultaneously.

(03.0)

Minimum Site Preparation Requirements

A smooth efficient installation is vital to Philips and their customers. Understanding what the minimum site preparation requirements are will help achieve this goal. The following list clearly defines the requirements which must be fulfilled before the installation can begin.

1. Walls to be painted or covered, baseboards installed, floors to be tiled and/or covered, ceiling shall have grid tiles and lighting fixtures installed and operational.

2. Doors and windows, especially radiation protection barriers, installed and finished with locksets operational.

3. All electrical convenience outlets, conduit, raceway and junction boxes installed and operational.

4. Incoming mains power operational and connected to main disconnect.

5. Room lighting installed and operational with contactor in circuit to permit foot pedal control. Devices should be level, parallel, and free of lateral or longitudinal movements.

6. 120V convenience outlets operational.

7. All support structures correctly installed. All channels, pipes, beams and/or other supporting devices should be level, parallel, and free of lateral or longitudinal movements.

8. All contractor supplied wires pulled and terminated.

9. A dust-free environment in and around the procedure room.

10. All HVAC (heating, ventilating and air conditioning) installed and operational as per specifications.

11. Architectural features such as computer floor, wood floor, casework, bulkheads, installed and finished.

12. All plumbing installed and operational.

13. Philips does not install or connect developing tanks, automatic processors or associated equipment, built in illuminators, cassette pass boxes, loading benches and cabinets, lead protective screens, panels or lead glass window and frame. This is to be done by the customer/contractor.

14. All doors and passage sizes are adequate for moving of equipment from exterior to specified room. Required door width is 48" (1220 mm) for 8'-0" (2440 mm) adjacent corridor and 72" (1828) for 6'-0" (1830 mm) adjacent corridor. Door heights shall be minimum 6'-8" (2030 mm) clear opening.

Note

Once Philips has moved equipment into the suite and started the installation, the contractor shall schedule his work around the Philips installation team on site. It is suggested that a telephone be provided in the room to receive telephone calls. This would alleviate facility staff from answering calls for Philips personnel.

Remote Service Diagnostics

Medical imaging equipment to be installed by Philips Medical is equipped with a service diagnostic feature which allows for remote and on site service diagnostics. To establish this feature, a RJ45 type ethernet 10/100/1000 Mbit network connector must be installed as shown on plan. Access to customer's network via their remote access server is needed for Remote Service Network (RSN) connectivity. All cost with this feature are the responsibility of the customer.

(02.0)

General Specifications

1. Responsibility

The customer shall be solely responsible, at its expense for preparation of site, including any required structural alterations. The site preparation shall be in accordance with plans and specifications provided by Philips. Compliance with all safety, electrical and building codes relevant to the equipment and its installation is the customer's responsibility. Sufficiency of such plans and specifications, specifically including, but not limited to the accuracy of the dimensions described therein, shall be the sole responsibility of customer. The customer shall advise Philips of conditions at or near the site which could adversely affect the carrying out of the installation work and shall ensure that such conditions are corrected and that the site is fully prepared and available to Philips before the installation work is due to begin. The customer shall provide all necessary plumbing, carpentry work, or conduit wiring required to attach and install products ready for use.

2. Permits

Customer shall obtain all permits and licenses required by federal, state/provincial or local authorities in connection with the construction, installation and operation of the products and shall bear any expense in obtaining same or in complying with any related rules, regulations, ordinances and statutes.

3. Radiation Protection

The customer or his contractor, at his own expense, shall obtain the service of a licensed radiation physicist to specify radiation protection.

4. Asbestos and Other Toxic Substances

Philips assumes no hazardous waste (i.e., pcb's in existing transformers) exists at the site. If any hazardous materials is found, it shall be the sole responsibility of the cusotmer to properly remove and dispose of this material at its expense. Any delays caused in the project for this special handling shall result in Philips time period for completion being extended by like period of time. Philips assumes that no asbestos material is involved in this project in any ceilings, walls or floors. If any asbestos material is found anywhere on the site, it shall be the customer's sole responsibility to properly remove and/or make safe this condition, at the customer's sole expense.

5. Labor

In the event local labor conditions make it impossible or undersirable to use Philips' regular employees for such installation and connection, such work shall be performed by laborers supplied by the customer, or by an independent contractor chosen by the customer at the customer's expense, and in such case, Philips agrees to furnish adequate engineering supervision for proper completion of the installation.

6. Schedule

The general contractor should provide Philips with a schedule of work to assist in the coordination of delivery of Philips supplied products which are to be installed by the contractor and delivery of the primary equipment.

7. Extended Installation or Turnkey Work by Philips.

Any room preparation requirements for Philips equipment indicated on these drawings is the responsibility of the customer. If an extended installation or turnkey contract exists between Philips and the customer for room preparation, then additional work required for the equipment will not be represented on these drawings. Some of the responsibilities of the customer as depicted in these drawings may be assumed by Philips. In the event of a conflict between the work described in the turnkey contract workscope and these drawings, the turnkey contract workscope shall govern.



- Customer Name -

City, ST

ACQSIM - CT System

Date

Drawn By

O.A. Number

Quote Number

Project Number

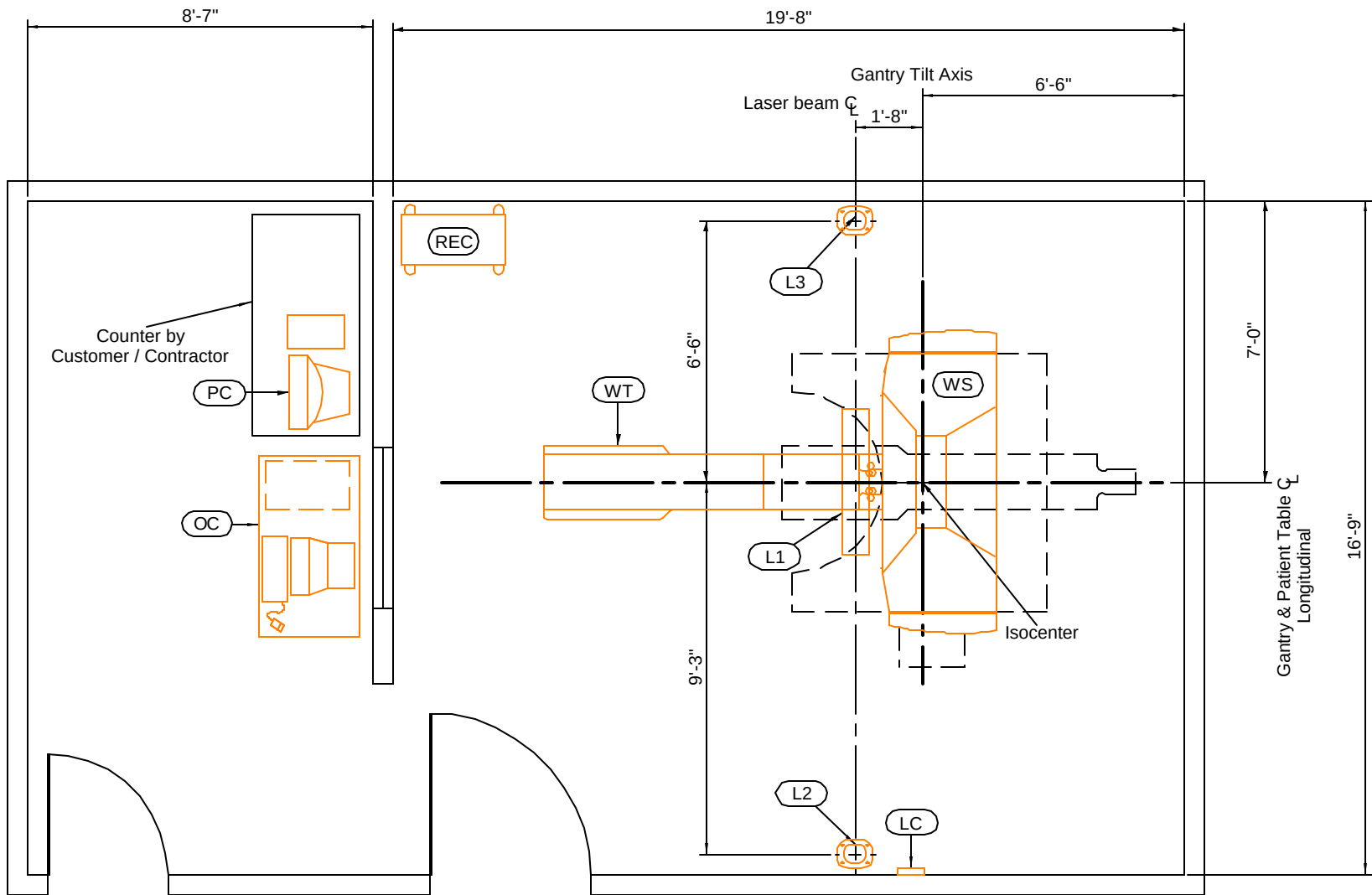
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AN

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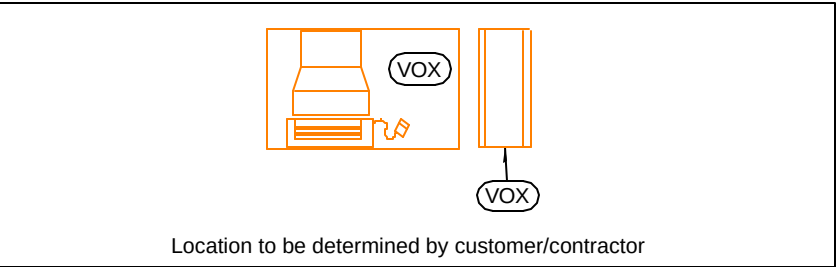


Equipment Layout

Recommended Ceiling Height : 9'- 0" (2743mm)



Equipment Legend				
A Furnished and installed by Philips B Furnished by customer/contractor and installed by customer/contractor C Installed by customer/contractor D Furnished by Philips and installed by contractor E Existing F Future G Option				
	Equipment Designation		Detail Sheet	
	Description		Weight (lbs)	Heat Load (btu/hr)
A	WS	ACQSIM CT Scanner Gantry	3400	AD1
A	WT	Patient Table	550	AD1
A	OC	ACQSIM CT Console	280	300 AD1
A	REC	ACQImage Recon Cabinet	147	1200 AD1
G	L1	Sagittal Laser Positioning Light	54	---
G	L2	CT-1 Side Laser Positioning Light (column mtd)	65	---
G	L3	CT-1 Side Laser Positioning Light (column mtd)	65	---
G	LC	Laser Light Controller	3	---
G	PC	Pinnacle Computer	32	---
G	VOX	Voxel-Q Workstation and Computer Module	350	5473



- Customer Name -
City, ST
ACQSIM - CT System

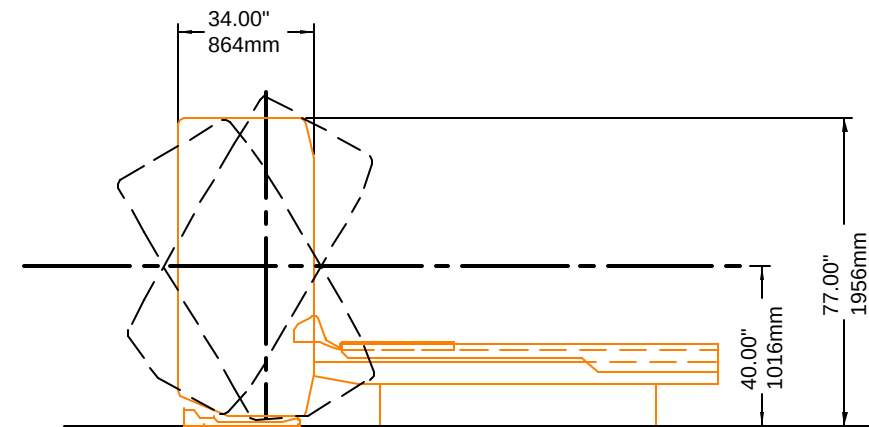
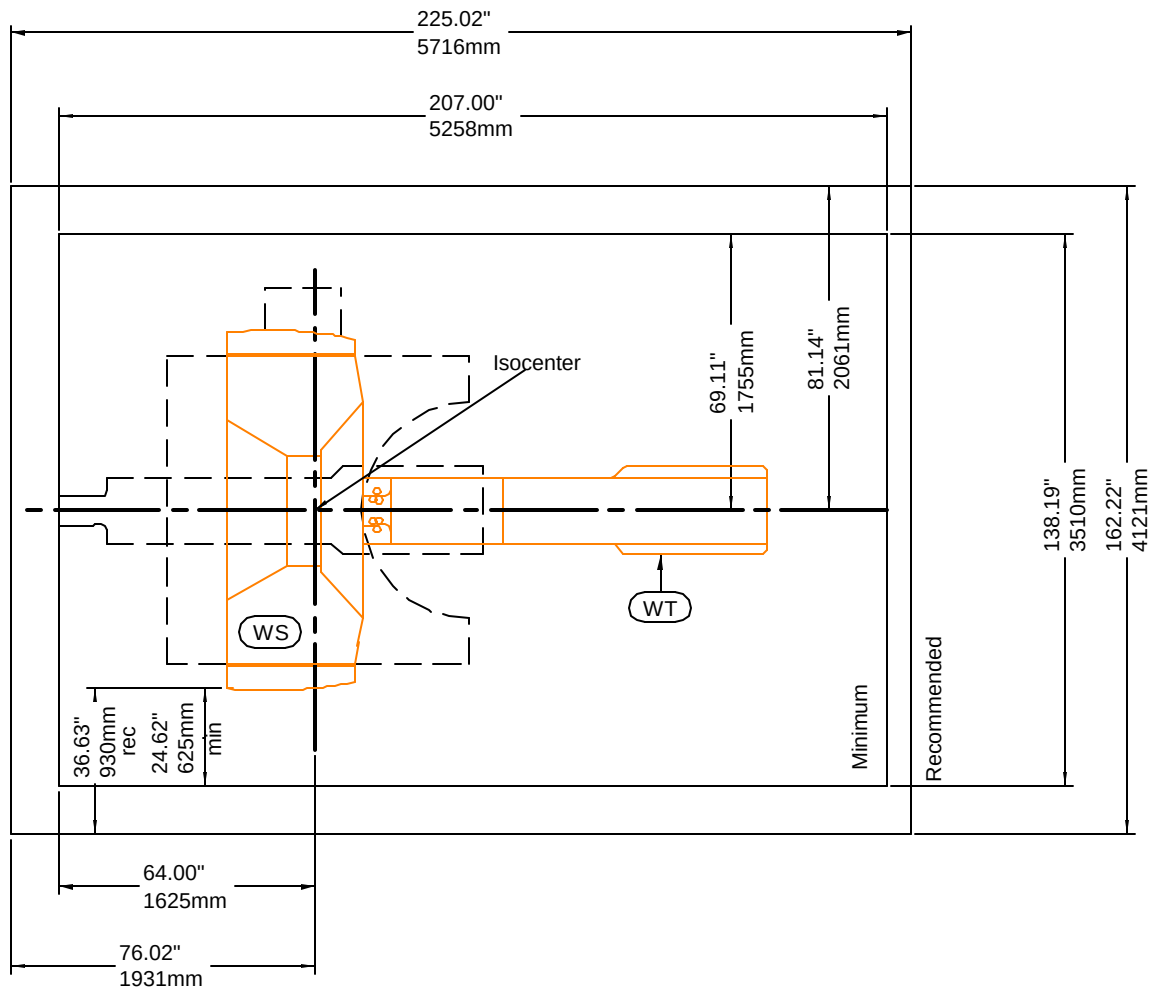
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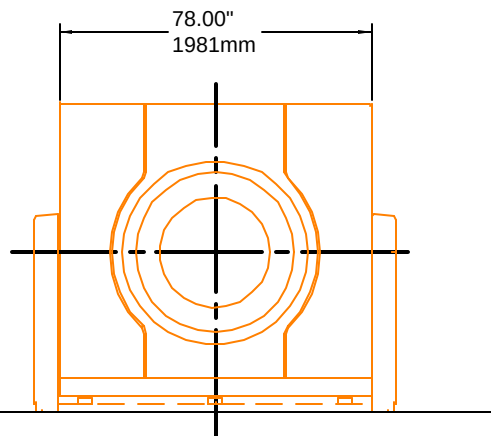
A1
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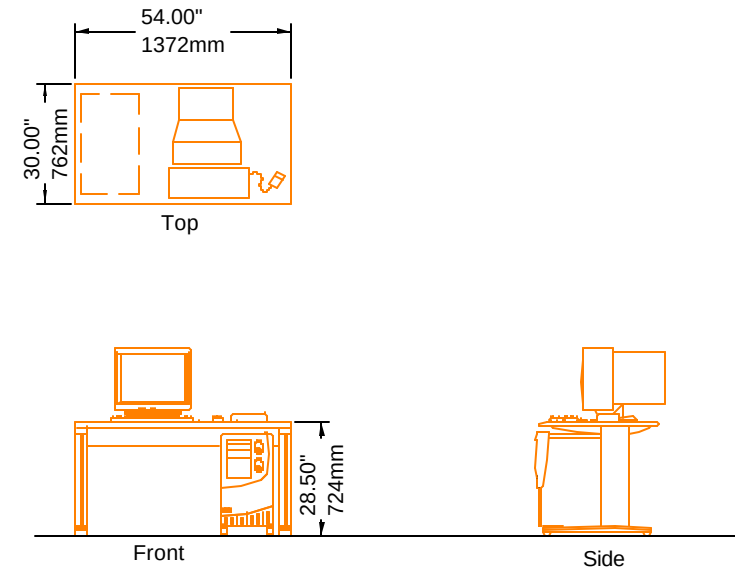
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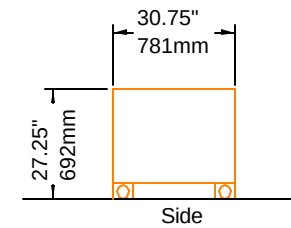
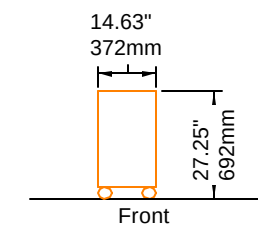
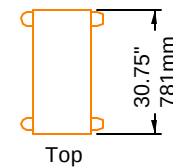
(03.0)		
WS		
ACQSIM CT Scanner Gantry		
Weight	3400 lbs	(1542 kg)
Heat Dissipation	21000 btu/hr	(5291 kcal/hr)



(03.0)		
WT		
Patient Table		
Weight	550 lbs	(249 kg)
Heat Dissipation	Included in gantry heat output	



(02.0)		
OC		
ACQSIM CT Console		
Weight	280 lbs	(126 kg)
Heat Dissipation	300 btu/hr	(75 kcal/hr)



(03.1)		
REC		
ACQImage Recon Cabinet		
Weight	147 lbs	(67 kg)
Heat Dissipation	1200 btu/hr	(302 kcal/hr)

- Customer Name -
City, ST
ACQSIM - CT System

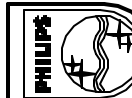
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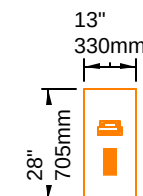
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AD2
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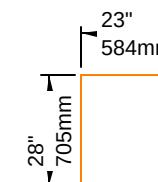
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[Top](#)



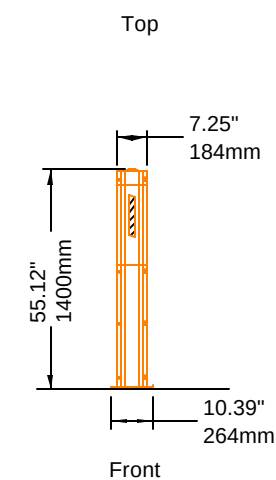
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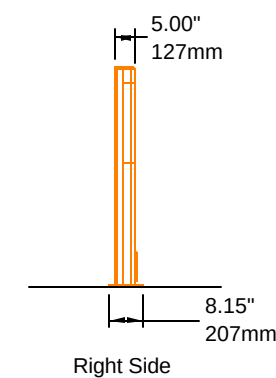
Side

(02.0)

SB	Sun Blade 2000 Workstation	
Weight	45 lbs	(20 kg)
Heat Dissipation	392 btu/hr	(98 kcal/hr)



Front

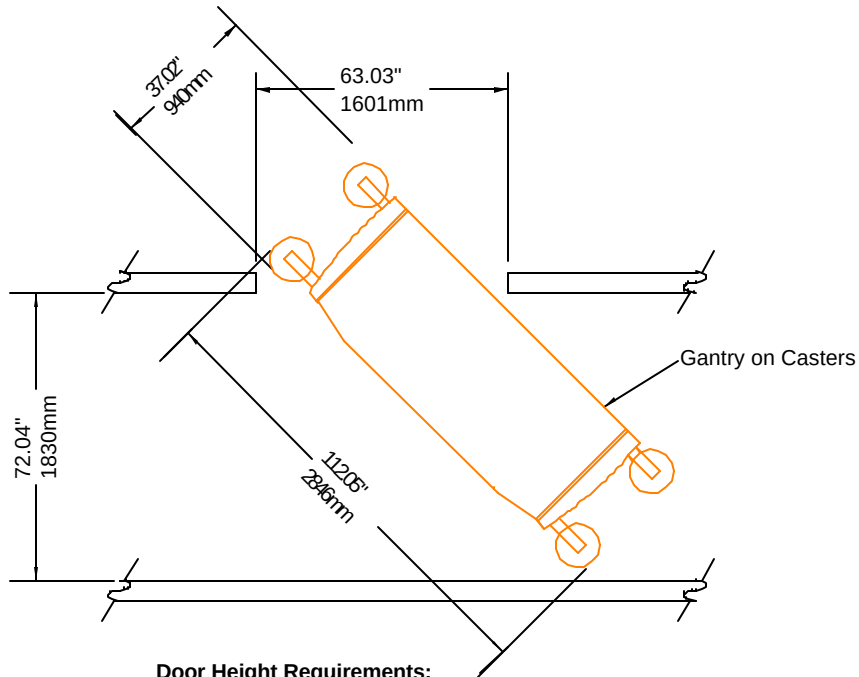


Right Side

(04.0)

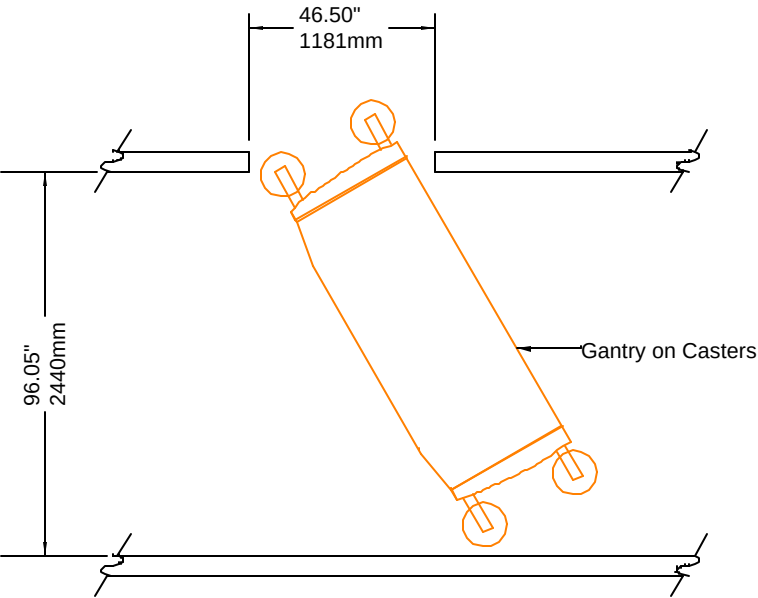
L2	L3	CT-1 Side Laser Positioning Light (column mtd.)
Weight	65 lbs	(29.5 kg)
Heat Dissipation	82 btu/hr	(20.7 kcal/hr)

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Door Height Requirements:

Scanner unit will fit through a standard 6'-8" (2032mm) door height with dolly assembly provided by Philips.

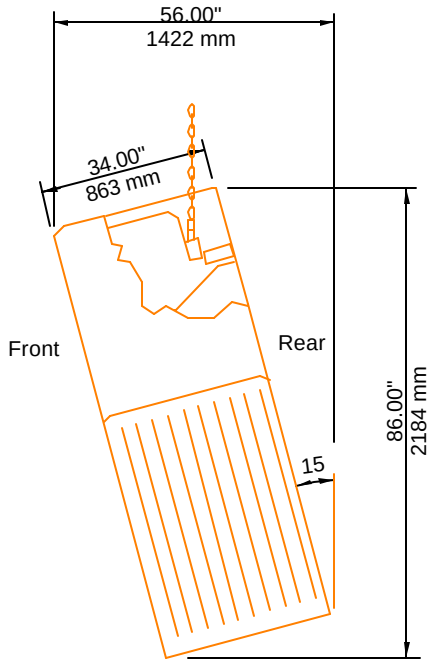


Note:

The casters have 8" (203mm) dia. hard plastic wheels. Tiled floors should be protected with 1/4" (6mm) thick tempered masonite panels or equivalent hard material.

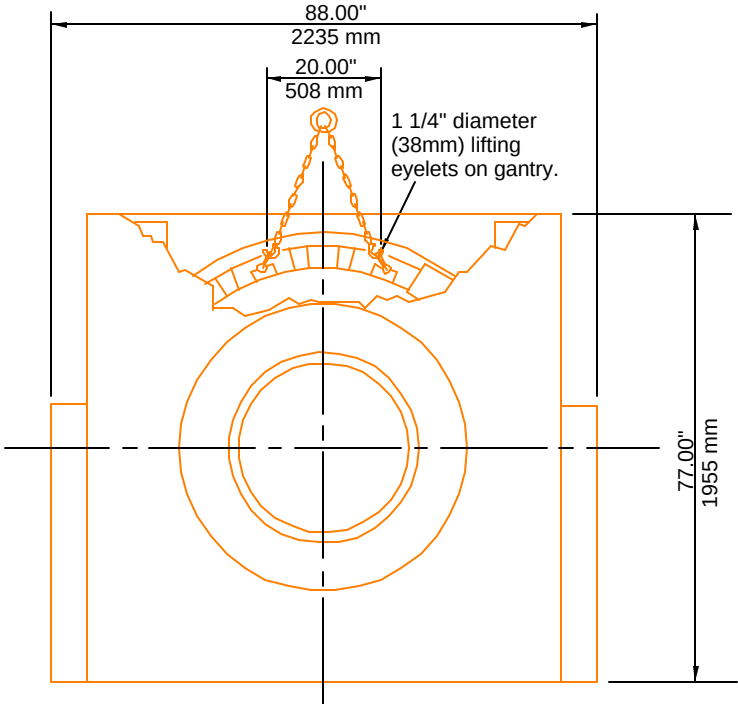
Shipping & Delivery Access

Not to Scale



Right Side View

Other lifting methods must be approved by Philips.



Rear View

Unit Weight: 3,400 lbs (1542 kg)

ACQSIM CT Scanner Gantry Lifting

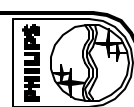
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ACQSIM Installation Notes

1. Floor anchoring
Contractor shall locate position of eleven floor anchors utilizing the ACQSIM CT floor template (order from the local Philips service representative; P.N. 174214). Contractor shall drill holes in floor to install "red head anchors" #S12 (rated pull-out force of 8,500 lbs. (37,808 N), or through bolting will be required (lead anchors are not acceptable). Contractor shall install eleven, 1/2" (12.7 mm) red head anchors, for scanner gantry and patient support, flush with finished floor surface.

In order to meet electrical ground isolation requirements, all methods of scanner gantry and patient support bolting must not come in contact with any building steel structure or reinforcement. Modular building floor structures may require additional bracing beneath scanner gantry and patient support, in order to eliminate vibration that can affect clinical images.

2. Environment
A sink with cabinet and counter space, etc., where required by code or desired by customer in each exam room. Consult with customer for requirements.

A toilet adjacent to CT room may be required by code or desired by customer. Consult with customer for requirements.

3. Safety
It is suggested that oxygen and suction services be provided in exam room convenient to gantry. Oxygen and suction may be portable or permanent type.

Fire protection for the type of equipment and systems specified shall be provided in accordance with local codes.

CT-1 Laser Lights Notes

1. All walls and ceiling structures shall be supplied and installed by the contractor to accommodate laser lights. Rigid mild steel support plates are required to support specified weight of each unit and assure rigid construction. It is important that vibration be kept to a minimum. See CT-1 laser light positioning and wall mounting details (SD2). Rigid structural supports installed from floor to ceiling structure of building are required if wall construction is inadequate to support mild steel support plates described. See additional notes below.

2. Philips service representatives shall verify precise positioning and alignment for each laser light, along with verification of supplied cable lengths based on proposed raceway/conduit routings specified on Philips site planning drawings.

3. All mounting hardware shall be provided by the contractor based on the type of wall support provided. Coordinate with local Philips service representative. C.M.U. block or poured concrete wall construction is preferred to meet rigidity requirements.

4. Levelness of floor and side wall/floor line locations shall match levelness under gantry within 3/16" (5mm).

5. Maximum distance from either side laser light to isocenter is 23'-0" (7010mm). Floor mounted side laser light support columns are available by Philips as an option when specified in sales contract, and are strongly recommended when gantry axis is not parallel to side walls.

- 6.** Sagittal laser light mounting recommendations:
- a. Preferred method: provide rigid ceiling support structure over patient couch area tied into building ceiling structure per Philips details and instructions, or;
 - b. Provide rigid floor to ceiling wall structure at a maximum distance of 20'-0" (6096 mm) from isocenter to mild steel support plate surface or;
 - c. Provide a rigid vertical soffit, within the 20'-0" (6096mm) maximum allowable distance range, extending from building ceiling structure. The soffit will simulate the wall mounting method. Apply this soffit method as last resort.

7. If building structure above false ceiling is inadequate to accommodate the ceiling mounted sagittal light mounting methods or floor to ceiling structure wall support columns to within 1 mm at isocenter, a free-standing bridge-type support frame shall be erected with vertical support columns attached to floor structure and independent from building ceiling structure.

8. Refer to additional Philips details for suggested surface or flush wall mounting methods for the side laser lights, which must be parallel to gantry axis. Philips strongly recommends that the side laser be protected from collision by mobile equipment or personnel.

9. Do not mount laser lights on sheet rock, drywall or suspended ceilings. Secure directly to rigid structure. Philips requires the use of mild steel support plates be supplied and installed on masonry walls or ceiling structure. Without rigidly mounted steel plate backing, stable isocenter positioning can not be guaranteed. The differential movement between the laser location and isocenter shall not exceed 1 mm. Philips shall mount laser lights after gantry isocenter has been established. Side laser lights may be rotated 90°at locations with wall obstructions below the lasers. For stud wall construction, steel support plates shall be welded to rigidly mount to floor and ceiling structures of the building. These structural members shall be independent of stud walls structure to avoid vibration transfer to the lasers.

Equipment Support Information

1. General
The customer shall be solely responsible, at its expense, for preparation of the site, including any required structural alterations. The site preparation shall be in accordance with this plan and specifications, the architectural/construction drawings and in compliance with all safety and building codes. The customer shall be solely responsible for obtaining all construction permits from jurisdictional authority.

2. Equipment Anchorage
Philips provides, with this plan and specifications, information relative to equipment size, weight, shape, anchoring hole locations and forces which may be exerted on anchoring fasteners. The customer shall be solely responsible, through the engineer of record for the building, to provide on the architectural/construction drawings, information regarding the approved method of equipment anchoring to floors, wall and/or ceiling of the building. Any anchorage test required by local authority shall be the customer's responsibility. Stud type anchor bolts should not be specified as they hinder equipment removal for service. Consult with Philips service prior to specifying anchor methods.

3. Floor Loading and Surface
Philips provides, with this plan and specifications, information relative to size, weight and shape of floor mounted equipment. The customer shall be solely responsible, through the engineer of record for the building, to provide on the architectural/construction drawings confirmation of the structural adequacy of the floor upon which the equipment will be placed. Any load test required by local authority, shall be the customer's responsibility. The floor surface upon which Philips equipment is to be placed/anchored shall be flat and level to within plus or minus 3/16 inch (4.8mm).

4. Ceiling Support Apparatus
Philips provides, with this plan and specifications, information relative to size, weight and shape of ceiling supported equipment. The customer shall be solely responsible, through the engineer of record for the building, to provide on the architectural/construction drawings, information regarding the approved method of structural support apparatus, fasteners and anchorage to which Philips will attach equipment. Any anchorage and/or load test required by local authority shall be the customer's responsibility.

Contractor to clearly mark Philips equipment longitudinal centerline on bottom of each structural support.

The structural support apparatus surface to which Philips equipment is to be attached, shall have horizontal equipment attachment surfaces parallel, square and level to within plus or minus 1/16 inch (2mm).

Any drilling and/or tapping of holes required to attach Philips equipment to the structural support apparatus shall be the responsibility of the customer.

Fasteners/anchors (i.e., bolts, spring nuts, lock and flat washers) and strip closures shall be provided by the customer.

5. Lighting
Lighting fixtures shall be placed in such a position that they are not obscured by equipment or its movement, nor shall they interfere with Philips ceiling rails and equipment movement or otherwise adversely affect the equipment. Such lighting fixture locations shall be the sole responsibility of the customer.

6. Ceiling Obstructions
There shall be no obstructions that project below the finished ceiling in the area covered by ceiling suspended equipment travel.

7. Seismic Anchorage (For Seismic Zones Only)
All seismic anchorage hardware, including brackets, backing plates, bolts, etc., shall be supplied and installed by the customer/contractor unless otherwise specified within the support legend on this sheet. Installation of electronic cabinets to meet seismic anchorage requirements must be accomplished using flush mounted expansion type anchor/bolt systems to facilitate the removal of a cabinet for maintenance. Do not use threaded rod/adhesive anchor systems. Consult with Philips regarding any anchor system issues.

8. Floor Obstructions/ Floor Coverings
There shall be no obstructions on the floor (sliding door tracks, etc.) in front of the Philips technical cabinets. Floor must be clear to allow cabinets to be pulled away from the wall for service.

Contractor to verify with Philips the preferred floor covering installation method.

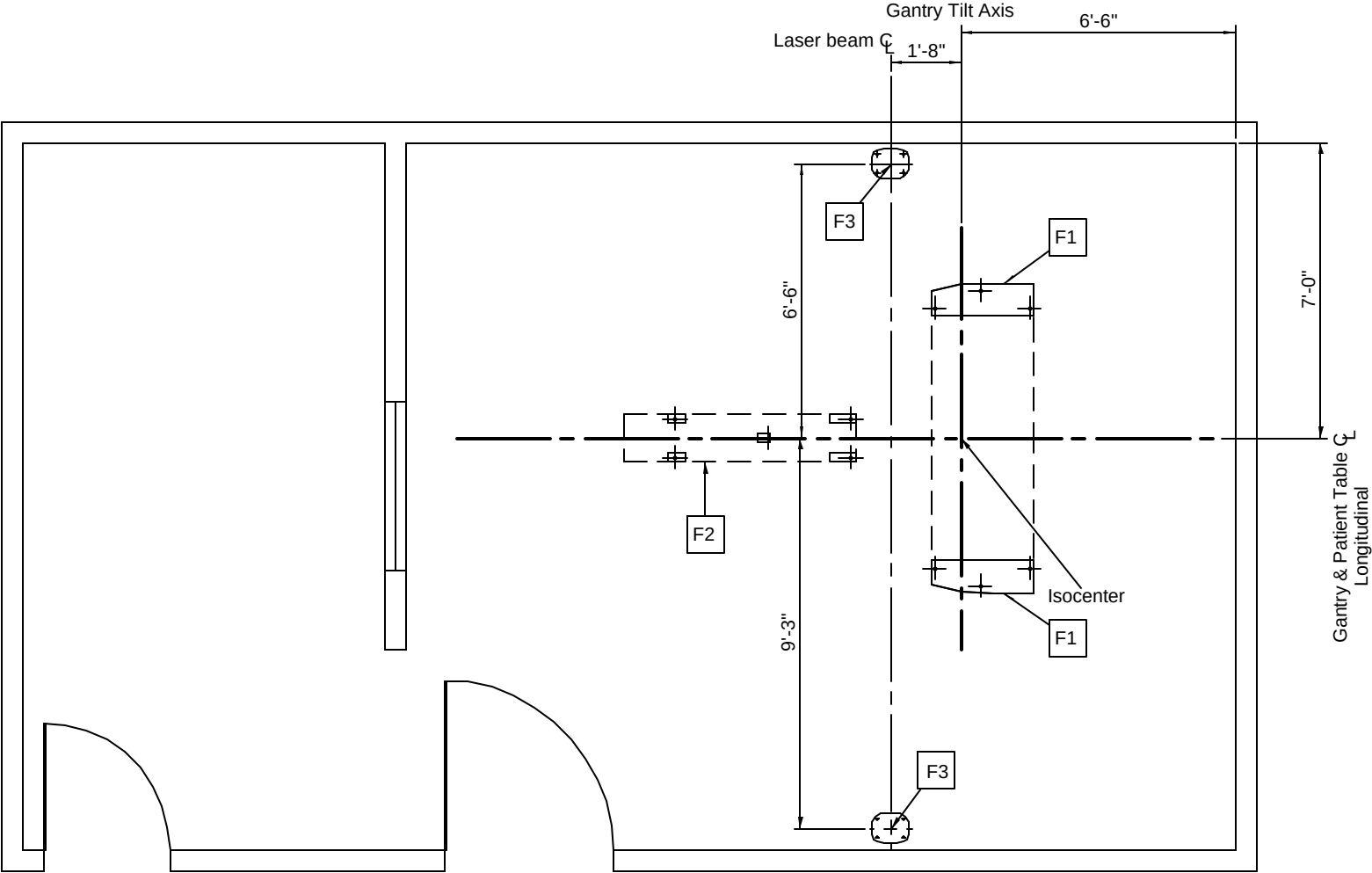
- Customer Name -
City, ST
ACQSIM - CT System

Drawn By	Date
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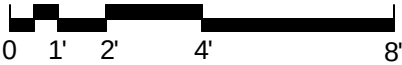
SN
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Floor Support Layout

Recommended Ceiling Height : 9'- 0" (2743 mm)



Floor Support Legend			
A B C D E F G			
		Item Number	Detail Sheet
		Description	
B	F1	Floor anchors for Gantry	SD1
B	F2	Floor anchors for Patient Table	SD1
B	F3	Floor anchorage for Side Positioning Laser Lights	SD2

- Customer Name -
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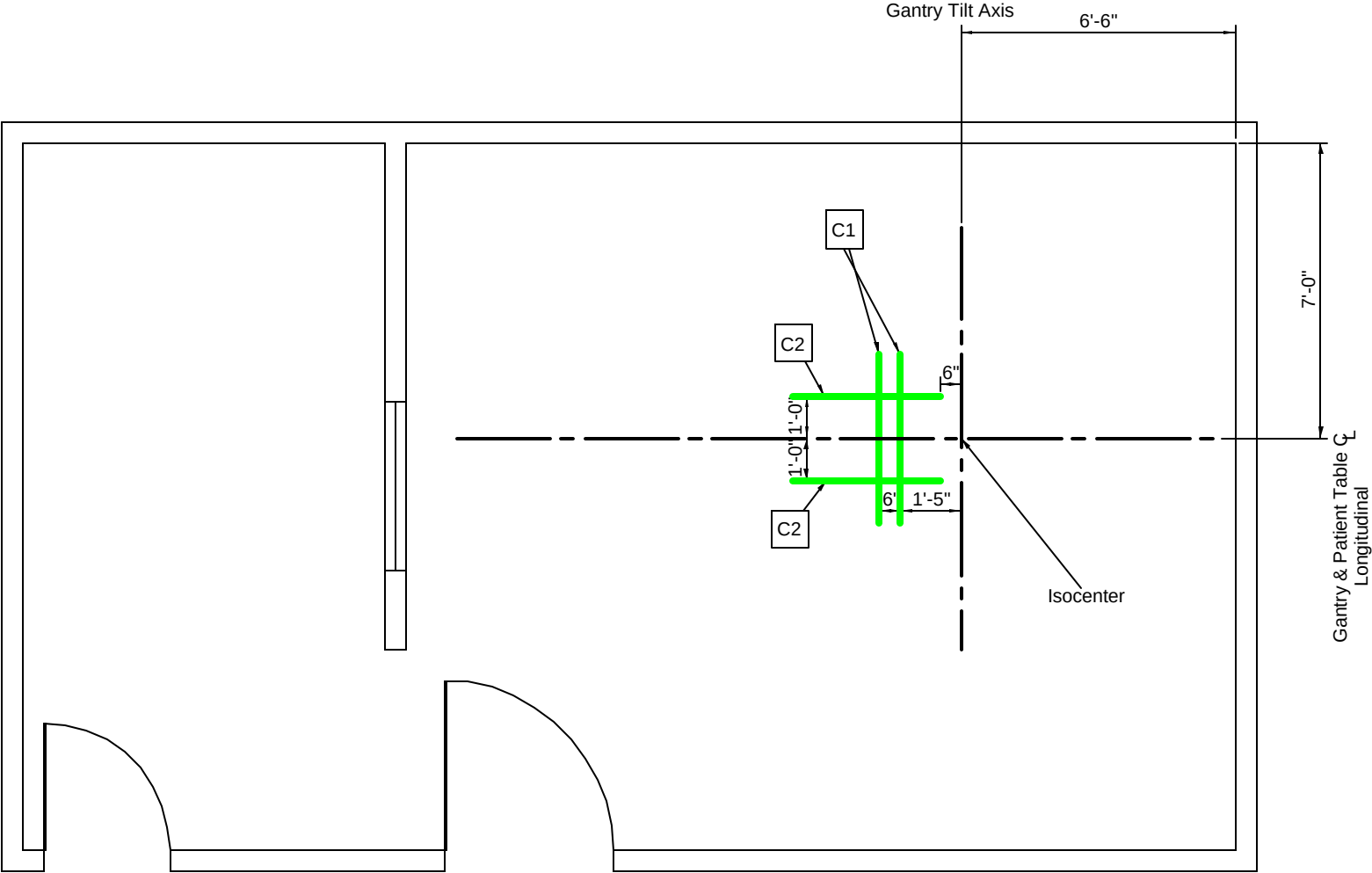
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S1
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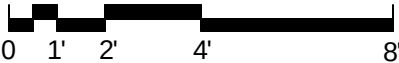
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06.01.05



Ceiling Support Layout

Recommended Ceiling Height : 9'- 0" (2743 mm)



Ceiling Support Legend			
A B C D E F G	Furnished and installed by Philips Furnished by customer/contractor and installed by customer/contractor Installed by customer/contractor Furnished by Philips and installed by contractor Existing Future Future		
	Item Number		Detail Sheet
	Description		
	B	C1	Adjustable strut channels
	B	C2	Fixed strut channels flush with false ceiling
			SD2
			SD2

- Customer Name -
City, ST
ACQSIM - CT System

Drawn By	Date
Quote Number	O.A. Number

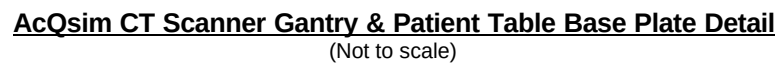
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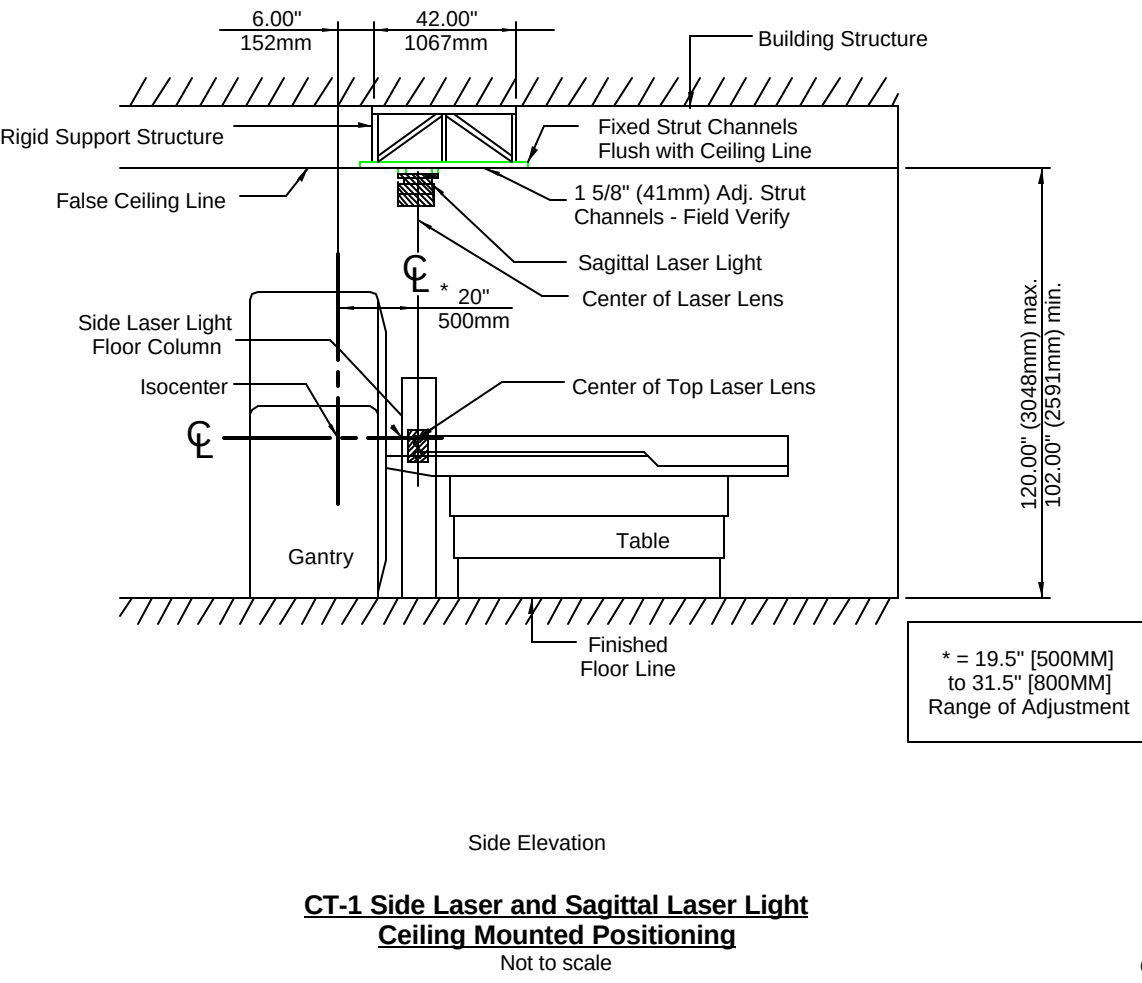
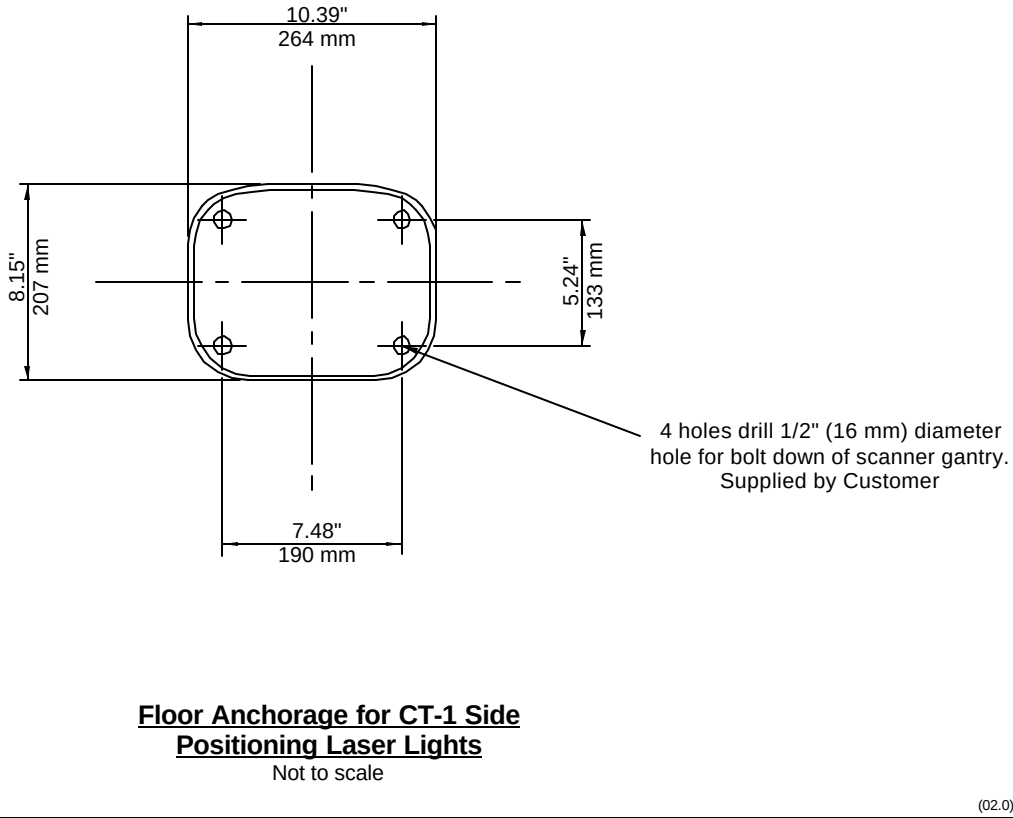
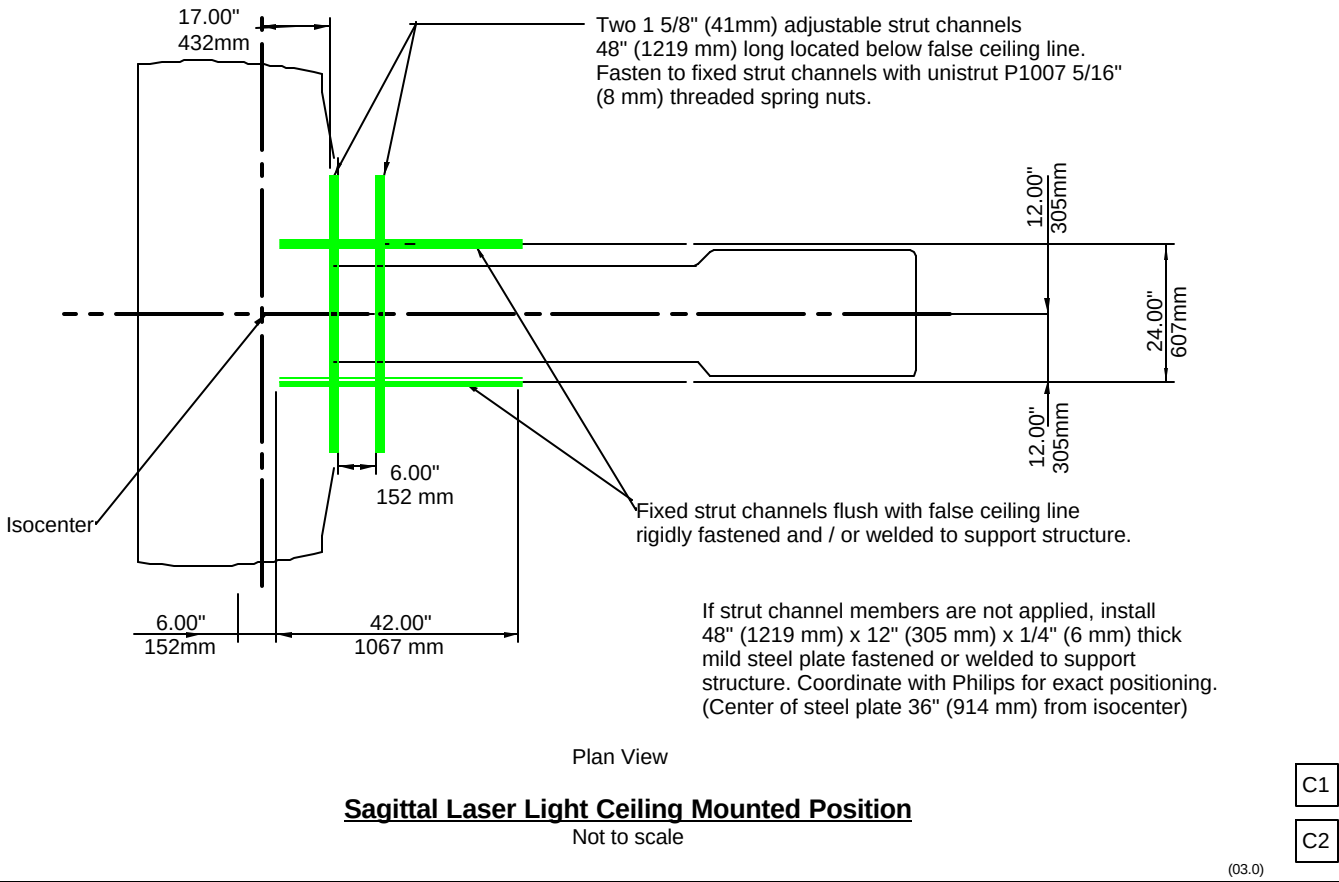


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- Customer Name -

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Project Number

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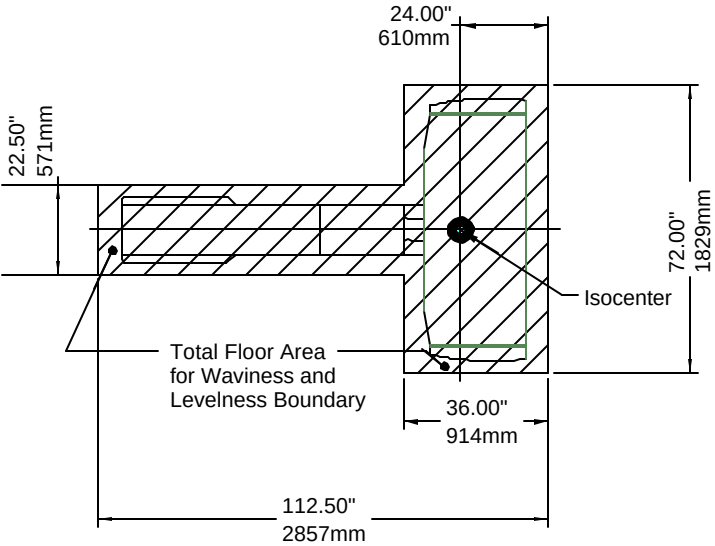
SD2

Sheet 11 of 20

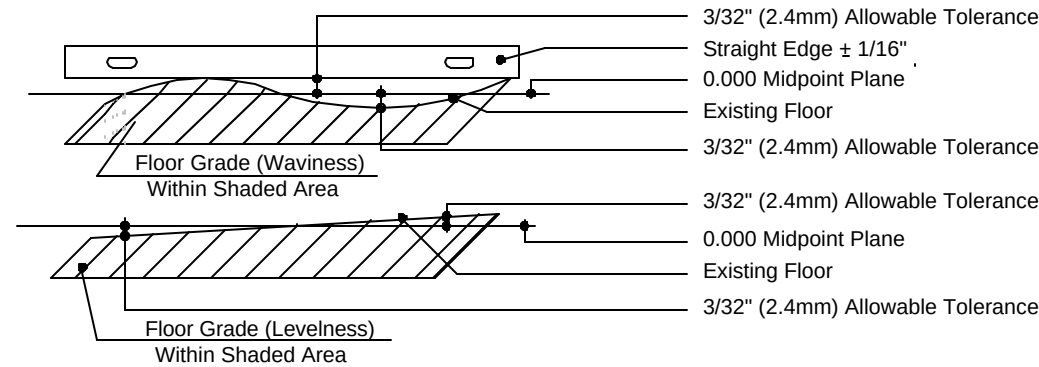
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NOTES:

1. Floor waviness of exam room shall not exceed 3/16" (4.8mm) when measured within 12" (300mm) from any point within the shaded area on layout.
2. Floor levelness of exam room shall not exceed 3/16" (4.8mm) when measured between any two boundary points across the shaded area on layout.
3. Finished floor coverings shall not be installed within shaded area. Gantry and couch base plates must be installed on structural concrete floor surface. Apply self leveling grout material as required to meet specifications.



Layout View



AcQsim CT Floor Levelness/Waviness Detail

Not to Scale

(05.0)



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06.01.05

General Electrical Information

1. General
The customer shall be solely responsible, at its expense, for preparation of the site, including any required electrical alterations. The site preparation shall be in accordance with this plan and specifications, the architectural/construction drawings and in compliance with all safety and electrical codes, the customer shall be solely responsible for obtaining all electrical permits from jurisdictional authority.

2. Materials and Labor
The customer shall be solely responsible, at its expense, to provide and install all electrical ducts, boxes, conduit, cables, wires, fittings, bushings, etc., As separately specified herein.

3. Electrical Ducts and Boxes
Electrical ducts and boxes shall be accessible and have removable covers. Floor ducts and boxes shall have watertight covers. Ducts shall be divided into as many as three separate channels by metal dividers, separately specified herein, to separate wiring and/or cables into groups as follows: Group a: power wiring and/or cables. Group b: signal and/or data and protective ground wiring and/or cables. Group c: x-ray high voltage cables. The use of 90 deg. ells is not acceptable. On ceiling duct and wall duct use 45 deg. bends at all corners. All intersecting points in duct to have cross over tunnels supplied and installed by contractor to maintain separation of cables.

4. Conduit
Conduit point - to - point runs shall be as direct as possible. Empty conduit runs used for cables may require pull boxes located along the run. Consult with Philips. A pull wire or cord shall be installed in each conduit run. All conduits which enter duct prior to their termination point must maintain separation from other cables via use of dividers, cross over tunnels, or flex conduit supplied and installed by contractor from entrance into duct to exit from duct. Maximum conduit lengths shown on these plans are calculated from electrical box entrance to electrical box entrance.

5. Conductors
All conductors, separately specified, shall be 75° c stranded copper, rung out and marked.

6. Disconnecting Means
A disconnecting means shall be provided as separately specified.

7. Grounding
Insulated grounding must conform with current requirements for electrically susceptible patient areas. See Article 517, National Electrical Code.

8. Warning Lights and Door Switches
"X-ray on" warning lights and x-ray termination door switches should be provided at all entrances to x-ray rooms as required by code.

9. Dimmer Switches
X-ray room lights should be provided with dimmer switches (including control room).

(02.0)

Electrical Notes

1. The contractor will supply & install all breakers, shunt trip and incoming power to the breakers. The exact location of the breakers and shunt trips will be determined by the architect or contractor.

2. The contractor shall supply & install all pull boxes, raceways, conduit runs, stainless steel covers, etc. Conduit/raceways must be free from burrs and sharp edges over its entire length. A greenlee pull string/measuring tape (part no. 435, or equivalent) shall be provided with conduit runs.

3. All pre - terminated, cut to length cables, will be supplied and installed by Philips Medical Systems. All cables to the breakers, will be supplied and installed by the contractor, subject to local arrangements.

4. Provide and install 4' - 2" (50 mm) dia. Chase nipples between adjacent wall boxes.

5. Electrical raceway shall be installed with removable covers. The raceway should be accessible for the entire length. In case of non - accessible floors, walls and ceilings, an adequate number of access hatches should be supplied to enable installation of cabling. Approved conduits may be substituted. All raceways will be designed in a manner that will not allow cables to fall out of the raceway when the covers are removed. In most cases, this will require above - ceiling raceway to be installed with the covers removable from the top. Raceway system as illustrated on this drawing are based upon length of furnished cables. Any changes in routing of raceway system could exceed maximum allowable length of furnished cables. Conduit or raceway above - ceiling must be kept as near to finished ceiling as possible.

6. Conduit sizes shall be verified by the architect, electrical engineer or contractor, in accordance with local or National Electrical Codes, whichever govern.

7. Convenience outlets are not illustrated. Their number and location are to be specified by the customer / architect. Locate at least one duplex outlet within 2'-0" (610 mm) of the system rack cabinet (s) and at least three (3) outlets spaced around the control room.

8. All sections of raceway and conduit shall be grounded with an independent #6 a.w.g.green wire that is to be attached using solderless lugs. All ceiling mounted structural support members and ceiling plates shall also be grounded. All grounding connections, terminals, etc. Shall be installed in a manner to provide accessibility for inspection, maintenance, repair, etc.

(03.0)

Electrical Legend

- A Furnished and installed by Philips
- B Furnished by customer/contractor and installed by customer/contractor
- C Installed by customer/contractor
- D Furnished by Philips and installed by contractor
- E Existing
- F Future
- G

Item Number		Detail Sheet	
		Description	
B	CB1	480 V, 3 phase, 3 pole, 100 amp circuit breaker. Location per local code or owner's requirements. Coordinate with PMSNA.	ED1 ED2 ED3
B	A1	120 V, single phase, single pole, remote ON / OFF class 9001 square D (or equal). BG210 surface mounted.	ED3
B	PDM	18" W x 24" L x 8" D junction box with blank, removable or hinged cover plate. Box to contain contractor supplied line monitor and magnetic contactor, with centerline 60" above finished floor.	ED3
B	WS	12" W x 12" L x 4" D junction box flush in finished floor. Box to tie into main trough system by means of a common grommated opening.	
B	L1	6" W x 6" L x 4" D junction box flush with false ceiling.	
B	L2 L3	4" W x 4" L x 4" D junction box flush in finished wall. Mounted at 12" above finished floor to bottom of box.	
B	LC	4" W x 4" L x 4" D junction box flush in finished wall. Mounted at 42" above finished floor to bottom of box.	
B	FR1	10" W x 3 1/2" D (or equivalent) floor trench duct flush mounted with water tight removable 1/4" thick steel cover plate. Provide barrier strip for separation of power wires and signal/data cables.	ED1
B	WR	10" W x 3 1/2" D (or equivalent) wall duct surface mounted with removable steel coverplate, 2" above finished floor. Provide grommated openings as needed. Verify with local Philips Service. Provide barrier strip for separation of power wires and signal/data cables.	ED1
B	R1	10"W x 3 1/2" D (or equivalent) riser surface mounted with removable steel coverplate. Riser to run from "FR1" up to "WR". Provide barrier strip for separation of power wires and signal/data cables.	ED1
B	REC OC	Grommated cable opening on "WR". Exact size and location to be determined by local Philips Service.	
B	PC	4" W x 4" L x 4" D junction box flush in finished wall. Exact location to be determined by local Philips Service.	
B		120 V/20 A dedicated duplex outlet for CT-1 Side Laser Positioning Lights Column Mounted (for optional upgrade to CT-3 side laser lights), Sagittal Laser Positioning Light, Pinnacle Computer, Sun Blade 2000 Workstation, and AcQsim CT Console.	
B	WL	Warning Light above X-ray room door, if required by code or desired by customer. Contractor to supply 120 V / 15A supply to warning light. (Optional - not shown).	ED2 ED3
B	DS	Door activated swith Square-D cat # class 9007 type AO (or equal), if required by code or desired by customer. Swith located on hinge side of door. (Optional - not shown).	ED2 ED3
B	AR	Install a ceiling air return intake vent in this area.	ED1
B	AF	Cool Air Feed - air flow across floor from rear wall vents required in this area.	ED1
B	S1	120 VAC / 15 A, single pole general purpose ON / OFF switch with red cover. Flush mounted 60" above finished floor to center line of box. Exact location to be determined by local Philips Service. Safety switch is required between scanner unit and "WL" for Philips Service. Locate adjacent to "CB1" when possible (not shown).	ED2 ED3
B	S2	120 VAC / 15 A, single pole Sagittal Laser Light ON / OFF switch. Flush mounted near "LC" (not shown).	
B	N	RJ45 type ethernet 10/100/1000 Mbit network connector. Access to customer's network via their remote access servier is needed for Remote Service Network (RSN) connectivity.	
B		120 V / 20 A dedicated duplex outlet (for Voxel-Q Workstation, not shown).	
B		Dedicated telephone jack (for Voxel-Q Workstation, not shown).	



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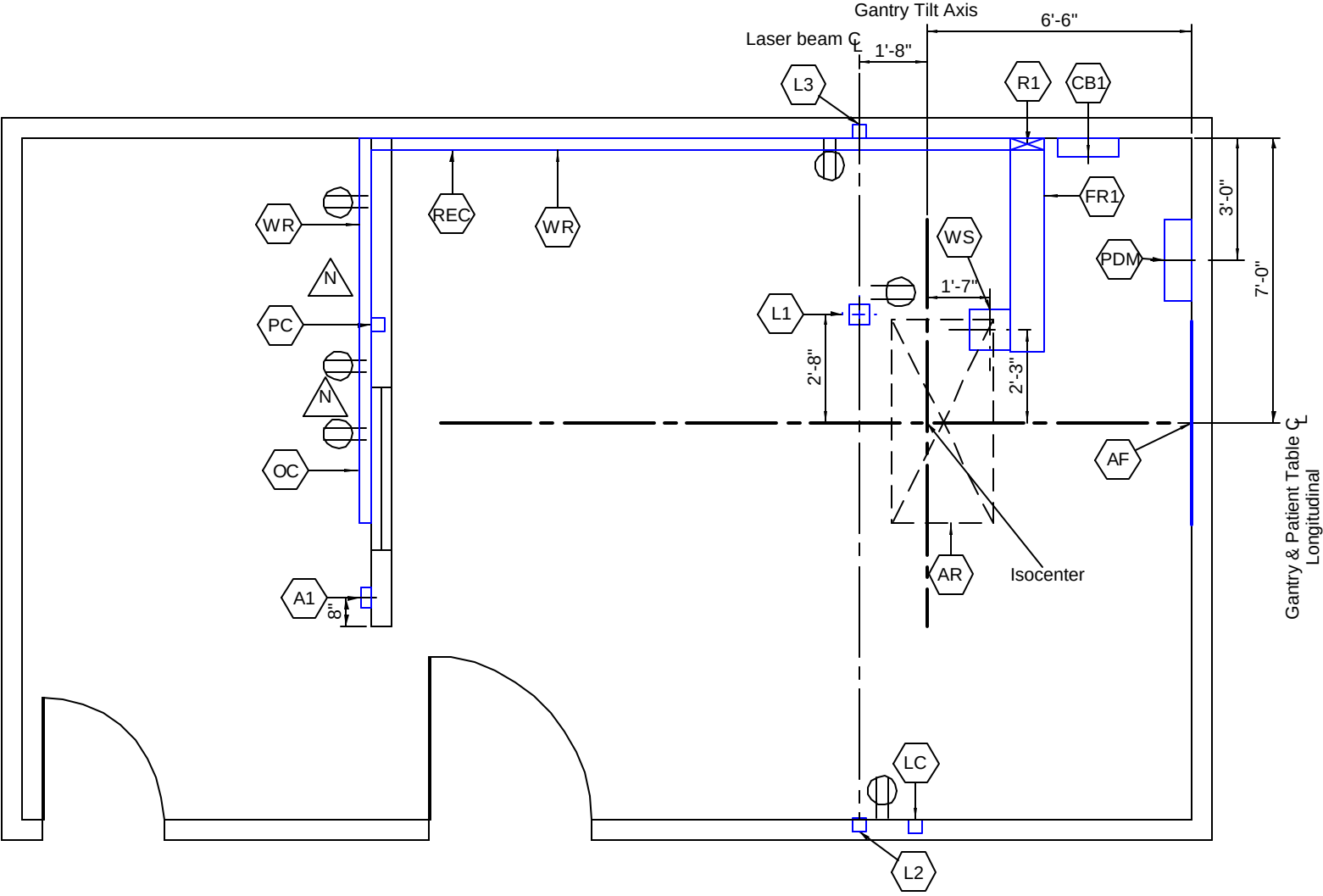
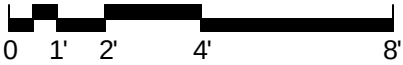
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Refer to Electrical Legend - Sheet EN
and Conduit - Sheet E2

Electrical Layout

Recommended Ceiling Height : 9'- 0" (2743 mm)



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Refer to Electrical Legend - Sheet EN
and Electrical Plan - Sheet E1

The use of 90 degree ells is not acceptable
use 45 degree bends at all raceway
corners. The use of crossover tunnels at
all applicable locations is required.

		Conduit Required					
General Notes							
1. All conduit runs must take most direct route point to point.							
2. All conduit runs must have a pull string.							
A Conduit supplied/installed by contractor - Philips cables installed by Philips							
B Conduit supplied/installed by contractor - Philips cables installed by contractor							
C Conduit and cables supplied and installed by contractor							
D Conduit existing - cables supplied and installed by Philips							
E Conduit existing - cables supplied by Philips, installed by contractor							
F Conduit existing - cables supplied and installed by contractor							
Conduit				Conduit Quantity	Minimum Conduit Size	Maximum Conduit Length	Special Requirements
Run No.	From	To					
C	1	Hosp. Power	WL	PER N.E.C.	PER N.E.C.	PER N.E.C.	The sum of run numbers 2 and 4 can not exceed 200'.
C	2	Hosp. Power	PDM	PER N.E.C.	PER N.E.C.	PER N.E.C.	
C	3	A1	PDM	1	1/2"	100	
C	4	PDM	CB1	1	2"	See Special Requirements	
C	5	CB1	WS	1	2"	50'	
C	6	WS	S1	1	1/2"	100'	
C	7	S1	WL	1	1/2"	100'	
C	8	WL	DS	1	1/2"	100'	
C	9	L1	S2	1	1/2"	50'	
B	10	LC	L1	1	1	50'	Run numbers 10, 11, 12, and 13 are plenum rated cables. Conduits maybe optional depending on local code or owner's requirements. Coordinate with local PMSNA.
B	11	L2	L1	1	1	50'	
B	12	L3	L1	1	1	50'	
B	13	PC	L1	1	1	50'	

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ACQSIM CT Electrical Specifications:

1. Distribution transformer shall be isolation type and a minimum of at least 75 kVA for ACQSIM CT, continuous rated, copper wound, with a maximum of 3% regulation, 2% preferable. Transformer shall be wye secondary feed with grounded neutral. No floating delta or corner / split leg grounding shall be applied. Reference N.E.C. Article 250-5 (d) and 250-26. Should the incoming primary source be stepped up by a customer supplied 208 V to 480 V transformer, the ACQSIM CT requirement will be a 112.5 kVA transformer to avoid the need of a power conditioner.
2. Philips Medical Services requires a line monitor with magnetic contractor drop out for power into C.T. system, in the event of severe low voltage or loss of phase condition. A manual reset button (ON) and shunt (OFF) in the control room is required. The contractor to supply and install:

Line monitor: SSAC Inc., Baldwinsville, NY. (315) 638-1300
o WVM 911 AL-60 (for ACQSIM - CT System)

Install on load side of power conditioner unit

Contractor: Square-D Co. (Worldwide Locations)
o Class 8903, SQG2VB1F4TX10 (for ACQSIM - CT System)

SQG2: 100 Amp, 3 Pole, NEMA 1 enclosure
F4T: Internal control transformer
V84: Control transformer - 208 V Primary / 120 V Secondary
V81: Control transformer - 480 V Primary / 120V Secondary
X10: Auxiliary contacts N.O.

Remote ON / OFF Switch: Square "D", Class 9001, BF 210 (flush mounted) or BG 210 (surface mounted)

Notice: When an optional power entrance controller box is supplied, the above line monitor and contractor do not have to be provided by the contractor. Also, as an option, the remote ON / OFF switch (A1) and circuit breaker (CB1) are available. Refer to Philips sales contract to determine whether or not these items are being supplied for each job. All of these items shall be installed by the contractor.

3. The scanner gantry and control console must not be in contact with any building structural steel, metal junction boxes, conduits and / or raceways. Reference N.E.C. Article 250-74 Excep. *4, 250-75 Excep., 384-20 Excep., 517-71, and N.F.P.A. Article 99, "Quiet Grounds."

4. Locate power conditioners, step-down transformers or isolation transformers close to the C.T. Suite when possible.

5. All power conditioning and surge suppressor equipment shall be installed according to manufacturer's specifications and installation instructions. Some devices may require additional external fuse protection. All work shall comply with local building codes.

6. **Important Notice:** Power supply for patient automatic power injector system should derive from a ground potential equal to the C.T. System.

7. Use a one-to-one isolation transformer, step-down isolation transformer or power conditioner wherever:

- a. The earth conductor is not available from the main power supply source; OR
- b. The run length of feeders exceeds 200' (61 m); OR
- c. It is necessary to improve power factors and reduce fault current; OR
- d. It is necessary to adjust voltages to meet equipment requirements; OR
- e. It is necessary to reduce harmonic currents induced in the supply.

Ground the newly derived neutral in accordance with N.E.C Article 250-26. However, do not use water piping, electrical conduits or electrical raceways as grounding electrodes. Also, performance of ground rods may be questionable due to environmental soil conditions at the site.

CB1

Electrical Requirement Notes

Electrical power distribution at the facility shall comply with:

Utilization voltages per ANSI C84.1 - 1982 range A.

Voltage to be supplied is 3 phase, grounded wye, 5 wire (3 phase, neutral and ground) unless otherwise noted in equipment specifications.

Phase conductors to be sized for instantaneous voltage drop per N.E.C. 517 - 73 and Philips recommendations.

Neural and ground conductors to be sized equivalently to phase conductors, unless otherwise noted.

Metal conduit shall not be used as the equipment ground conductor.

Clamping type surge suppressors are highly recommended in addition to standing facility lighting arrestors. Equipment to be protected from ANSI/EEE C62.41-1980 location category B impulses.

ANSI / NFPA 70 - National Electrical Code
Article 250 - grounding
Article 517 - health care facilities

ANSI / NFPA 99 - health care facilities

NEMA standard XR9 - power supply guideline for x-ray machines

Power Quality Guidelines

1. Power supplied to medical imaging equipment must be separate from power feeds to air conditioning, elevators, outdoor lighting, and other frequently switched or motorized loads. Such loads can cause waveform distortion and voltage fluctuations that can hinder high quality imaging.

2. Equipment that utilizes the facility power system to transmit control signals (especially clock systems) may interfere with medical imaging equipment, thus requiring special filtering.

3. The following devices provide a high impedance, nonlinear voltage source, which may affect image quality:
Static UPS systems, Series filters, Power conditioners, and Voltage regulators.

Do not install such devices at the mains supply to medical imaging equipment without consulting Philips installation or service personnel.

4. Line impedance is the combined resistance and inductance of the electrical system and includes the impedance of the power source, the facility distribution system, and all phase conductors between the source and the imaging equipment. Philips publishes recommended conductor sizes based on equipment power requirements, acceptable voltage drops, and assumptions about the facility source impedance. The minimum conductor size is based on the total line impedance and N.E.C. requirements. Unless impedance calculations are performed by an electrical engineer, the recommended values must be used.

FR1

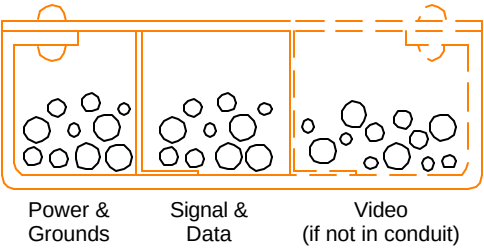
R1

WR

Cable Trough Divisions

Troughs or ducts must be separated by metal barriers into three sections:

1. Power cables and ground cables can be run together.
2. Signal cables and data cables can be run together but must be separated from power cables.
3. Video cables to be run separately from all other cables.

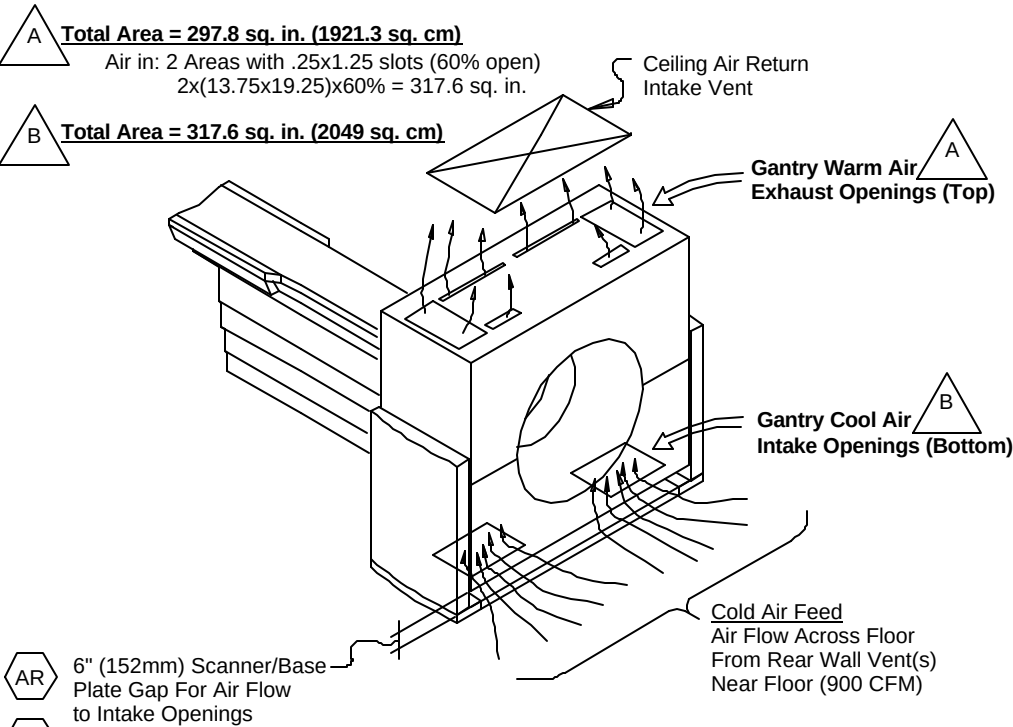


5. It is important that all cables are placed in the appropriate trough and at no given point do any cables from one division cross with cables from another. Trough separation must be continuous from the beginning to the end of the run.
6. Trough or ducts: steel with steel dividers grounded to building ground.
7. Contractor to provide cable restraints in all troughs.

(02.0)

Air Out: 6 Areas with .25x1.25 Slots (60% Open)

2x(10.25x18.75)x60% = 230.6 sq. in. (1487.7 sq. cm)
2x(2.75x10.25)x60% = 33.8 sq. in. (218.1 sq. cm)
2x(1.25x22.25)x60% = 33.4 sq. in. (215.5 sq. cm)



AR

AF

ACQSIM CT Gantry Air Flow Detail

Not to Scale

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ED1

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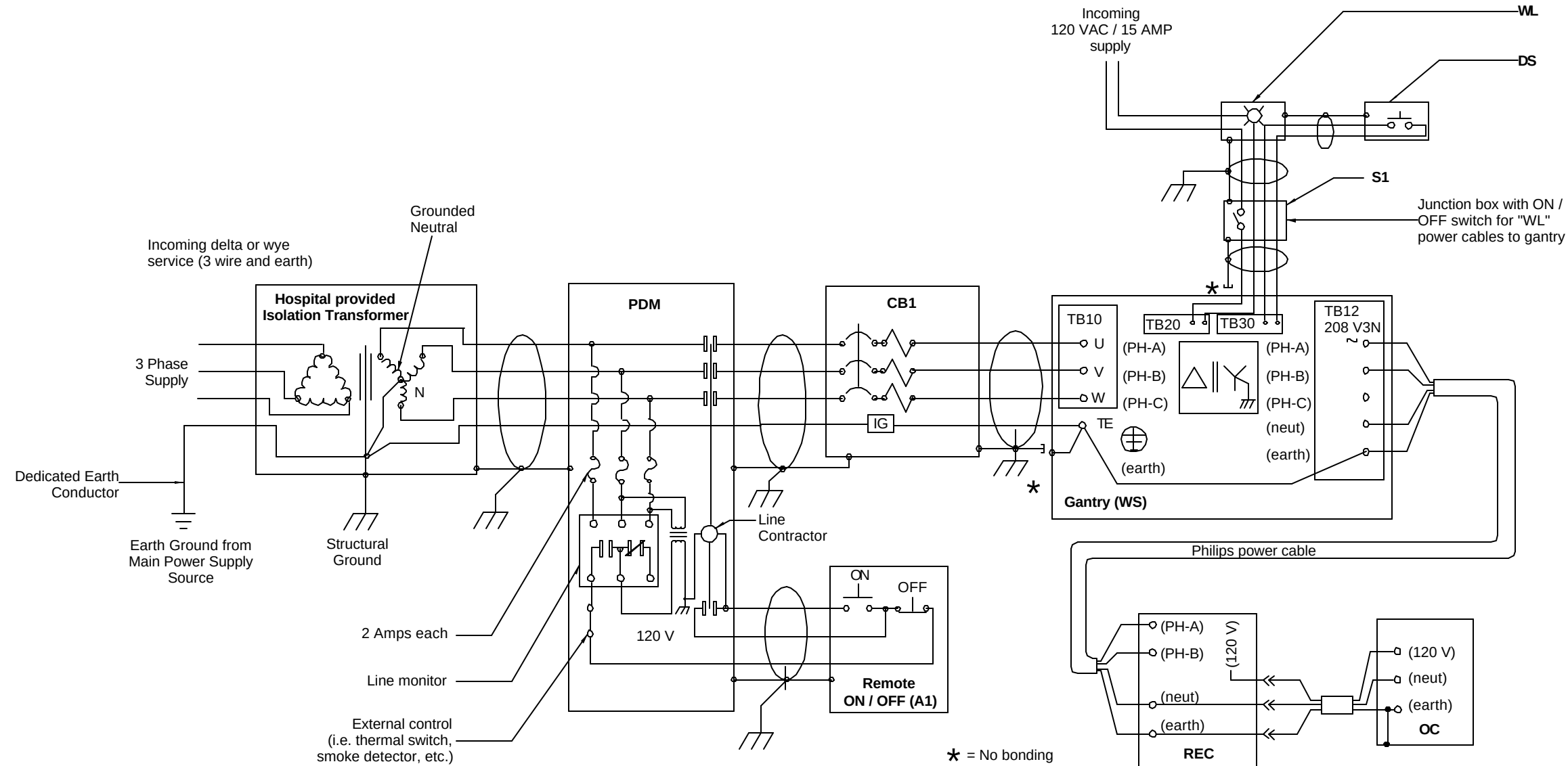


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(02.0)

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ACQSIM CT Incoming Power Schematic



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06.01.05

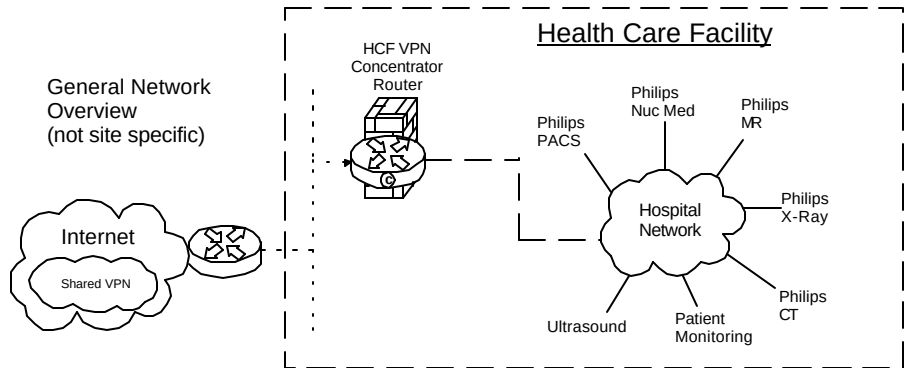
Philips Medical Systems Remote Services Network (RSN)

Broadband Site-to-Site Connectivity (Recommended)

This connectivity method is designed for customers who prefer a connection from the RSN Data Center to the Health Care Facility (HCF) utilizing their existing VPN equipment.

Connectivity Details:

- A Site-to-Site connection from the RSN data center's Cisco router will be established to the HCF's VPN concentrator
- The VPN Tunnel will be an IPSEC, 3DES encrypted Tunnel using IKE as standard, but alternative standards are also available, such as AES, MD5, SHA, Security Association lifetime and Encryption Mode
- Every system that we will be servicing remotely will have a static NAT IP that we configure on the RSN Data center side.



Action Required by Hospital:

- Review and approve connection details
- Complete appropriate Site Checklist
- Configure and allow Site-to-Site access prior to setting up connectivity depending on the access criteria that the HCF decides to implement (ex: Source IP filtering, destination IP filtering, NAT assignment, etc.)
- Route traffic from within the hospital network with destination addresses 192.68.48.0/22 to the designed IP provided by Philips

Broadband Router Installed at Health Care Facility

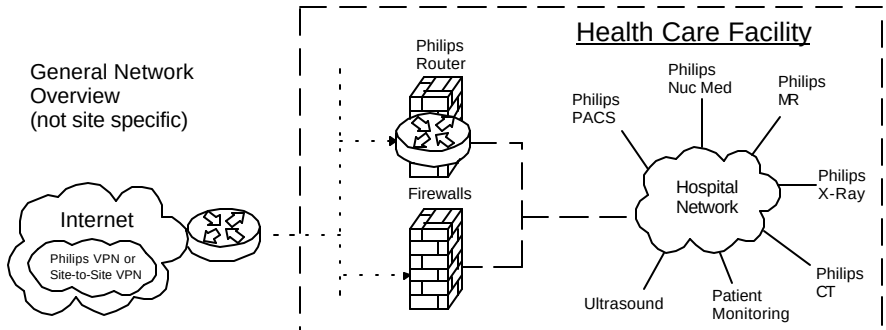
This connectivity method is designed for customers who have a dedicated high speed connection for Philips equipment.

Connectivity Details:

- An RSN Cisco 1711 or 1712 router will be preconfigured and installed at the HCF by Philips in conjunction with the HCF IT representative.
- The VPN Tunnel will be an IPSEC, 3DES encrypted Tunnel using IKE and will be established from the RSN-DC and terminated at the RSN Router on-site
- One to One NAT is used to limit access to Philips equipment only
- Router Config and IP auditing is enabled for Customer IT to view via website 24/7
- Dedicated DSL connections are also supported

Option 1: Parallel to HCF Firewall Connectivity Method

This connectivity method is designed for customers who prefer a Philips RSN Router installed on site utilizing all the security features provided and managed by Philips.

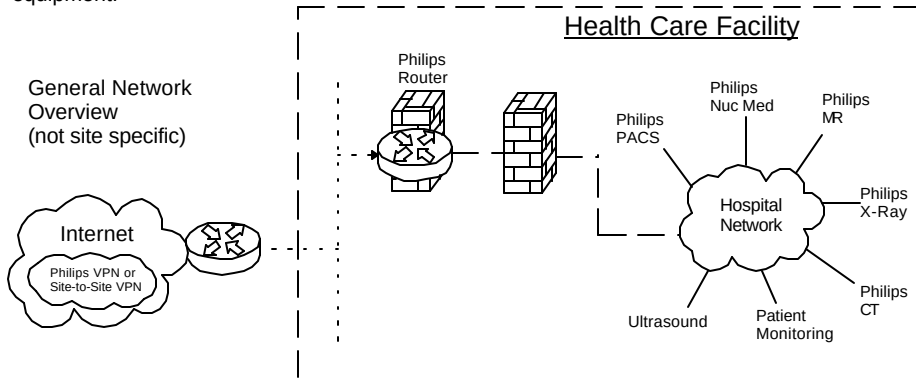


Action Required by Hospital:

- Assign a fixed public IP Address from the ISP to be configured on the Philips router. This is the DOTTED link on the picture connected to the firewall.
- Assign a Back end IP for the Philips router on the Hospital Network
- Complete appropriate Site Checklist
- Route traffic from within the hospital network with destination addresses 192.68.48.0/22 to internal Philips router Ethernet interface. This is the DASHED line connected to the firewall.

Option 2: Back End Connected to the HCF Firewall Connectivity Method

This connectivity method is designed for customers who prefer a Philips RSN Router installed on site by setting up an IP-Based policy allowing access thru existing HCF Firewall to Philips equipment.

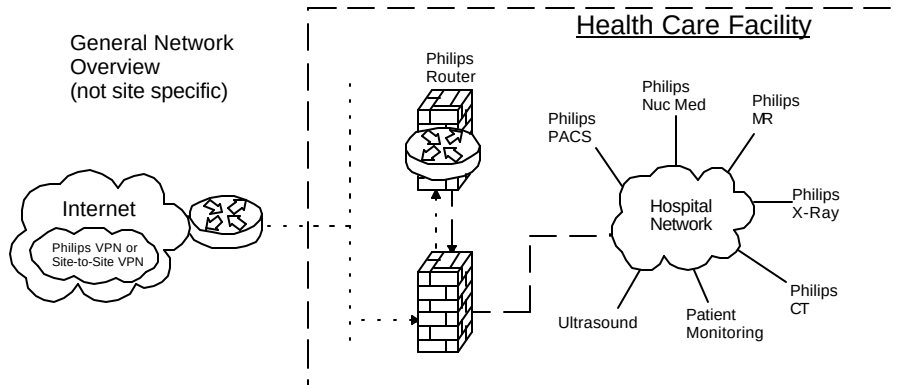


Action Required by Hospital:

- Assign a fixed public IP Address from the ISP to be configured on the Philips router. This is the DOTTED link on the picture connected to the firewall.
- Assign a Back end IP for the Philips router on the Hospital Network
- Complete appropriate Site Checklist
- Route traffic from within the hospital network with destination addresses 192.68.48.0/22 to internal Philips router Ethernet interface. This is the DASHED line connected to the firewall.
- Configure and allow on the firewall on the DASHED line interface access between the IP address allocated by the hospital to the Philips internal Ethernet router interface and the target modality IP address.

Option 3: Router Installed Inside the HCF's DMZ

This connectivity method is designed for customers who prefer the RSN Router installed inside and existing, or new DMZ, allowing access to Philips equipment.



Action Required by Hospital:

- Assign a fixed public IP Address from the ISP to be configured on the Philips router. This is the DOTTED link on the picture connected to the firewall.
- Assign a Back end IP for the Philips router on the Hospital Network
- Complete appropriate Site Checklist
- Route traffic from within the hospital network with destination addresses 192.68.48.0/22 to internal Philips router Ethernet interface. This is the DASHED line connected to the firewall.
- Configure and allow on the firewall on the DASHED line interface IPsec protocol communication by opening protocol 500, 50, 51, 47 and port 23 + TACACS. Traffic should be between external IP Address located on the Philips router and the RSN Data center IP address 192.68.48/24 and IP address AOSN TACAS
- Configure and allow on the firewall on the DASHED line interface access between the IP address allocated by the hospital to the Philips internal Ethernet router interface and the target modality IP address

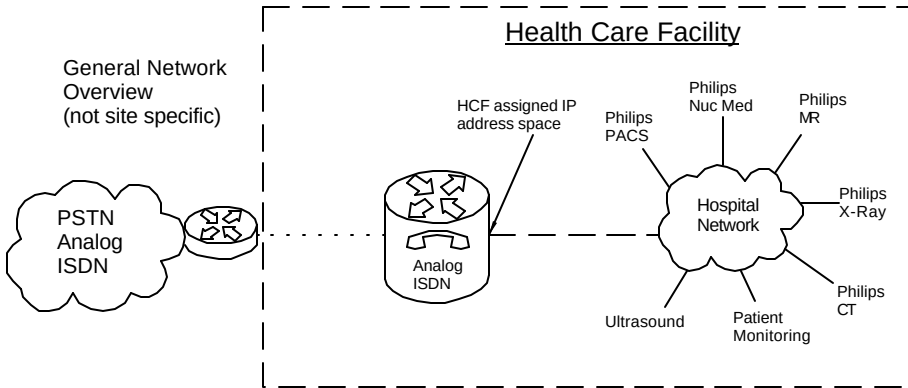
Dialup Router Installed at Health Care Facility

Dedicated Dial-Up Connectivity Method

This connectivity method is designed for customers who do not have internet access or who prefer a dedicated Analog or dedicated ISDN connection.

Connectivity Details:

- This connection method is meant for customers that do NOT have High Speed access to the Internet
- A Cisco 1721 router with an Analog or ISDN interface will be installed at the customers facility
- The RSN data Center DSR (dial out server) will establish a connection to the router on a site using CLI and CHAP. A NAT table will be configured on the router to limit access to only specific and allowed Philips systems.



Action Required by Hospital:

- Assign a fixed, dedicated Analog or ISDN phone line
- Assign an internal IP address for the Philips router on the HCF network
- Complete appropriate Site Checklist
- Route traffic from within the hospital network with destination addresses 192.68.48.0/22 to internal Philips router Ethernet interface.

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Quote Number
O.A. Number

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Instructions

This form is to be used by Project Manager, Contractor and Service Engineer.

Information is used to develop and determine site ready date.

Items listed are go/no go items for delivery unless noted as delay only items.

Items identified with *** as delayed items must be completed after hours or on weekend. These itmes cannot be accomplished while installation is in progress. Also, these items must be completed within two days of installation start or they may stop installation.

Site Readiness Checklist

Modality: _____

OA#: _____

Site Name: _____

Location: _____

Contact Name: _____

Contact Phone Number: _____

- ☐ Customer site preparation verified in general against the Philips final planning drawings.
- ☐ Walls finished including painting.
- ☐ Doors installed.
- ☐ Floor leveled according to Philips drawings and specifications.
- ☐ Floors are tiled/covered finished. Flooring is covered with protective covering (scratch protection).
- ☐ Ceiling lights installed.
- ☐ Cable conduit and ductwork installed and clean. Position checked. Duct covers in place but not finally closed. Cable opening are clear, without sharp edges. Pull strings in conduit. Installation per Philips specifications.
- ☐ HVAC environmental equipment installed and working according to Philips specifications.
- ☐ Ceiling installation completed.
- ☐ Electrical preparation according to Philips specifications.
- ☐ All network cabling, drops installed according to Philips specifications (including hardcopy cameras).
- ☐ All pre-cabling identified on Philips drawings has been installed.
- ☐ Pre-move survey completed - Delivery route identified.
- ☐ Led glass installed ***.
- ☐ X-ray warning lights installed ***.
- ☐ Dedicated phone line for modem use***.
- ☐ Room has been cleaned ***.
- ☐ Cabinets and casework installed***.
- ☐ RSN Surveys completed and submitted

Approved for Delivery

Project Manager

Date

Service Engineer

Date

- Customer Name -
City, ST
ACQSIM - CT System

Drawn By

Date

Quote Number

O.A. Number

Project Number

N-SRD-02002

CHK

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